

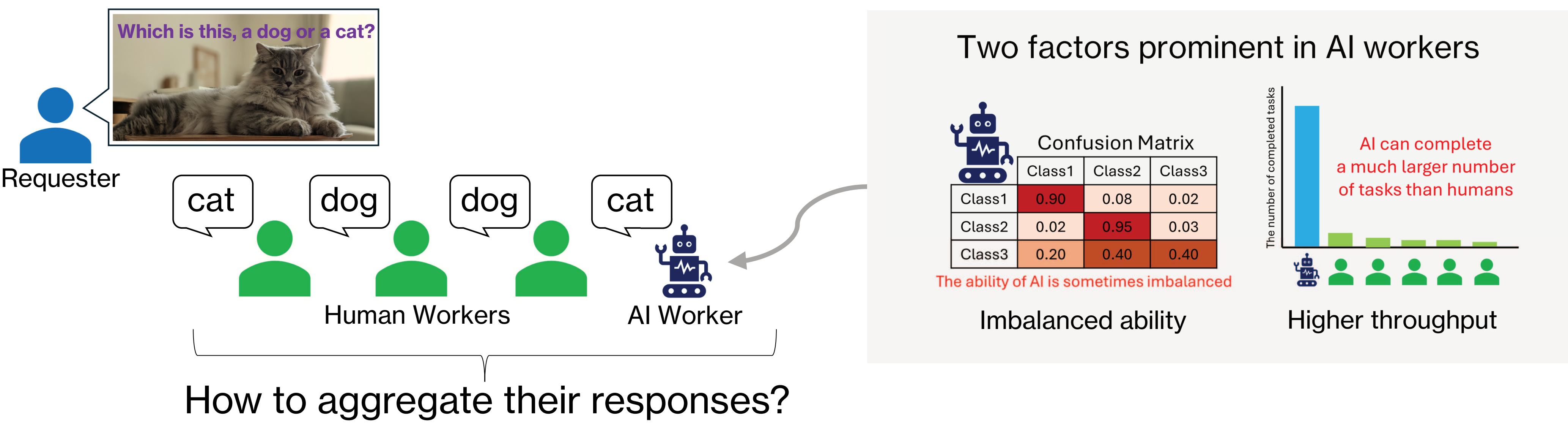


Paper



GitHub

The Difference Between Humans and AIs Affects Aggregation Results



Dawid-Skene (DS)

Latent Variables:

$$\pi = \begin{bmatrix} \pi_{11} & \dots & \pi_{1c} \\ \vdots & \ddots & \vdots \\ \pi_{c1} & \dots & \pi_{cc} \end{bmatrix}$$

Worker's Confusion Matrix

OneCoin model

Latent Variables:

$$a$$

Worker's Ability

GLAD

Latent Variables:

$$\alpha$$

Worker's Ability

$$\beta$$

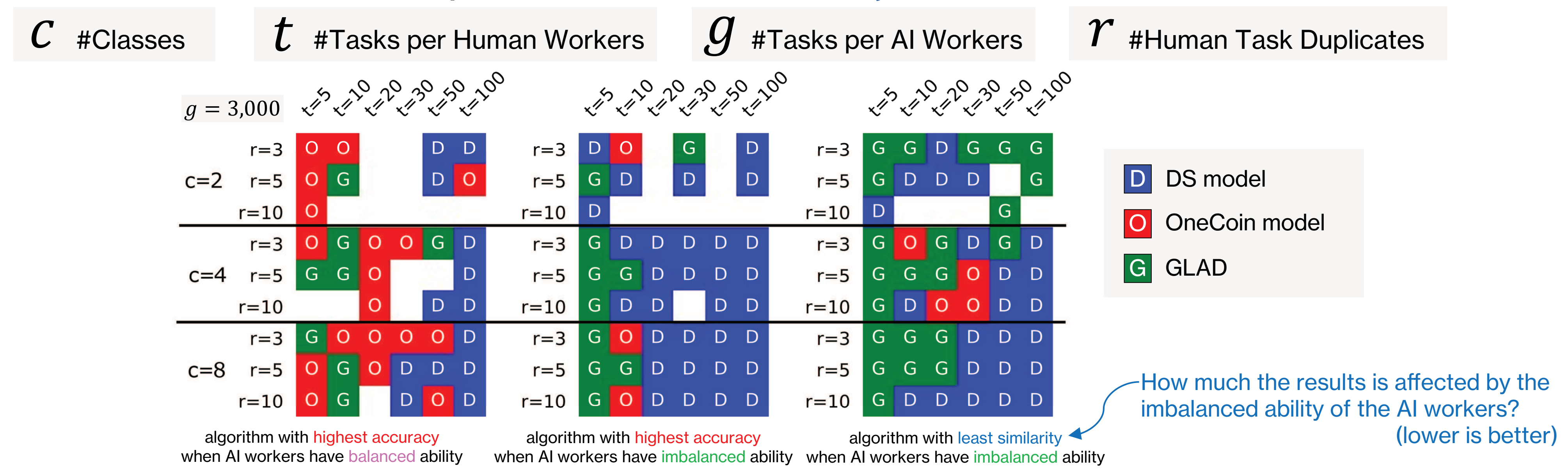
Task's Difficulty

Existing aggregation algorithms based on the EM algorithm can estimate workers' ability

RQ1: **What are the key factors to affect the quality of aggregation results?**

Answer: 1. The number of tasks per human workers
2. Balance or Imbalance of AI worker's ability

Settings in our simulation: We experimented with a total of 1,026 cases.



The DS model is better when AI workers have imbalanced ability, except when t is small.

Hypothesis → Is this due to the difficulty estimating confusion matrices of human workers?

RQ2: **What algorithm should we use considering the key factors?**

Answer: Hybrid algorithm achieves higher accuracy and mitigates AI's bias

We propose

DS+OneCoin model

The **hybrid algorithm** of the DS and OneCoin model that attempts to estimate confusion matrices well when t is small

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Worker's Confusion Matrix

$$a$$

Worker's Ability

